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FIITJEE

JEE(Main)-2016

ALL INDIA INTEGRATED TEST SERIES

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST–VII
(MAIN)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	D	B
2.	A	B	B
3.	B	B	C
4.	A	B	C
5.	C	C	A
6.	C	B	A
7.	B	A	B
8.	B	D	A
9.	C	D	D
10.	B	C	C
11.	B	A	D
12.	C	D	A
13.	C	A	B
14.	C	B	A
15.	A	A	C
16.	C	C	B
17.	B	D	C
18.	B	A	B
19.	D	C	C
20.	D	C	A
21.	D	C	D
22.	D	C	C
23.	C	A	C
24.	A	B	C
25.	A	A	A
26.	A	B	A
27.	A	C	A
28.	C	B	C
29.	C	A	C
30.	D	C	A

Physics

PART – I

Hints & Solutions

1. The velocity of P along AP equals that of A along AP.

2. Force by liquid = Mg
$\Rightarrow$ $\rho(Av)v = Mg$
But $Av = 1 \times 10^{-3}$ m^3/s and
$v = \sqrt{(20)^2 - 2\times10\times15} = 10$ m/s
$\therefore$ M = 1 kg

3. The motion of the particle within the tunnel is simple harmonic.

4. Total mechanical energy = gravitational P.E. + K.E., total mechanical energy is conserved.

5. $dQ = n\left(\frac{\gamma R}{\gamma - 1}\right)dT$, $dU = \frac{nR}{\gamma - 1}dT$
dW = nRdT
dQ : dW : dU
$\Rightarrow$ $\gamma : \gamma - 1 : 1$

6. Using equation of Trajectory, we get
$u = 15\sqrt{2}$ m/s

8. From the free body diagram, we get,
$N = mg\cos\alpha + ma_0\cos\alpha$
$mg\sin\alpha + ma_0\sin\alpha = ma$
$a = (a_0 + g)\sin\alpha$
$L = \frac{1}{2}(a_0 + g)\sin\alpha t^2$
$t = \sqrt{\frac{2L}{(a_0 + g)\sin\alpha}}$

9. $F\cos\theta = \mu N$ .....(A)
$N = F\sin\theta + mg$ .....(B)
By (A) and (B)
$F\cos\theta = \mu(F\sin\theta + mg)$
$F\cos\theta - \mu F\sin\theta = \mu mg$
$F(\cos\theta - \mu\sin\theta) = \mu mg$
if F is ∞. Hence $\cos\theta - \mu\sin\theta = 0$
$\cos\theta - \mu\sin\theta = 0 \Rightarrow \cot\theta = \mu$
$\Rightarrow$ $\theta = \cot^{-1}(\mu)$

10. Stress = Y × strain
force $F = Y\left(\frac{A\Delta\ell}{\ell}\right)$
$\Rightarrow$ $F = \pi r^2 Y\left(\frac{A\Delta\ell}{\ell}\right)$

11. $\ell_2 - \ell_1 = (\ell'_0 - \ell_0) + t(\ell'_0\alpha_2 - \ell_0\alpha_1)$
By the condition given in the question,

$\ell_2 - \ell_1 = \ell'_0 - \ell_0$

$\therefore \quad t(\ell'_0\alpha_2 - \ell_0\alpha_1) = 0$

$\ell'_0 = 60$ cm

12. $f_{max} = md\omega_{0_{max}}^2$

$\omega_{0_{max}} = \sqrt{\frac{\mu mg}{md}} = \sqrt{\frac{\mu g}{d}}$

13. $y = x$

$PE = \frac{1}{2}m\omega^2y^2$

$KE = \frac{1}{2}m\omega^2(a^2 - y^2)$

$TE = PE + KE = \frac{1}{2}m\omega^2a^2$

14. $T = Kx = m\omega^2r$
where $r = \ell_0 + x$

$\therefore \quad Kx = m\omega^2(\ell_0 + x) \Rightarrow \quad x = \frac{m\omega^2\ell_0}{K - m\omega^2}$

$\therefore \quad r = \ell_0 + x = \frac{K\ell_0}{K - m\omega^2}.$

15. $V_{max} = (2\pi f)\, A$
$a_{max} = (2\pi f)^2 A$
$\frac{a_{max}}{V_{max}} = (2\pi f).$

16. Acceleration $\propto$ - displacement

17. In second harmonic of an organ pipe pressure antinode will be at $\frac{\lambda}{4}$ from the either side of the open organ pipe and $\lambda = L$ (length of the organ pipe).

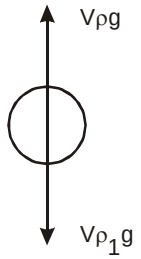
18. $W_{agent} = \Delta K + \Delta U$
for minimum work done $\Delta K = 0$
$W_{agent} = \Delta U = mgR$

19. Use the definition of C.M. :
$x_{cm} = \frac{\Sigma m_i x_i}{\Sigma m_i}$

20. From the F.B.D. of the body

$a = \frac{V\rho_1 g - V\rho g}{V\rho_1} = \left(1 - \frac{kh}{\rho_1}\right)g \qquad \because \rho = kh$

$\therefore \quad v\frac{dv}{dh} = \left(1 - \frac{kh}{\rho_1}\right)g$



$\Rightarrow \quad \frac{v^2}{2} = \left(h - \frac{k}{2\rho_1}h^2\right)g$

$\therefore \quad v = 0$ at $h = \frac{2\rho_1}{k}$

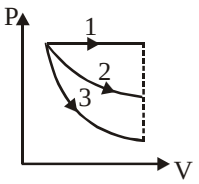
also $a = 0 \Rightarrow h = \frac{\rho_1}{k}$ is the mean position.

$\therefore \quad$ amplitude $= \frac{\rho_1}{k}$

21. Calculate the field due to a semi-circular ring and then use superposition principle.

22. Total mechanical energy = K.E. of satellite + P.E. of satellite.

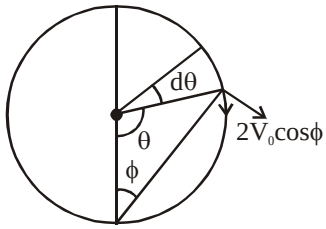
23. In the diagram since adiabatic process curve for P-V diagram is steeper than isothermal P-V curve. So curve 3 will denote adiabatic process, curve 2 isothermal and curve 1 isobaric. i.e. the minimum area enclosed with v-axis will be for adiabatic process



24. $a = -(dv/dt)$
$kv^2 = -(dv/dt)$,
$v = v_0/(1 + kv_0t)$

25. $a_R = \frac{|\vec{v}|^2}{R}$

27. $dm = \left(\frac{m}{2\pi R}\right)(Rd\theta) = \frac{md\theta}{2\pi}$



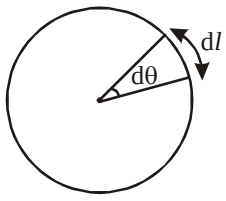
or, $V = 2V_0\cos\phi = 2V_0\sin\frac{\theta}{2}$

$dK = \frac{1}{2}(dm)\times 4V_0^2\times\sin^2\frac{\theta}{2}$

or, $dK = \frac{mV_0^2}{2\pi}(1-\cos\theta)d\theta$

$K_{AOB} = \frac{mV_0^2}{2\pi}\int_{-\pi/2}^{+\pi/2}(1-\cos\theta)d\theta = \frac{mV_0^2}{2} - \frac{mV_0^2}{\pi}$.

28. $2T\sin\frac{d\theta}{2} = dma_R = dmR\omega_0^2$



.....(A)
Mass per unit length = mass/length

$dm = \frac{m}{2\pi}d\theta$

By equation (A) $2T\sin\frac{d\theta}{2} = dmR\omega_0^2$

$T = \frac{mR\omega_0^2}{2\pi}\left\{\sin\frac{d\theta}{2} = \frac{d\theta}{2}\right\}$

Chemistry

PART – II

SECTION – A

5. $\underset{(Z)}{B} + \underset{}{F_2} \longrightarrow \underset{(X)}{BF_3}$

$8BF_3 + 6LiH \longrightarrow \underset{(Y)}{B_2H_6} + 6LiH_4$

6. $n_1 + n_2 = 4$, $n_2 - n_1 = 2$

$n_1 = 1$, $n_2 = 3$

$$\bar{\nu} = Rz^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R \times 2^2\left[\frac{1}{(1)^2} - \frac{1}{3^2}\right] = \frac{32R}{9}$$

16. $\text{Step}: 1 \;\; 2NO \rightleftharpoons N_2O_2$

$\text{Step}: 2 \;\; N_2O_2 + H_2 \longrightarrow N_2O + H_2O(\text{slow})$

$\text{Step}: 3 \;\; N_2O + H_2 \longrightarrow N_2 + H_2O$

Rate = $K'[N_2O_2]\,[H_2]$

From step 1 $K_c = \dfrac{N_2O_2}{[NO]^2} \Rightarrow [N_2O_2] = K_c[NO]^2$

$\text{Rate} = K'.K_c[NO]^2[H_2] = K[NO]^2[H_2]$

17. Let number of moles of $Ca_3(PO_4)_2$ and H_3PO_3 are x and y respectively

$$\frac{x}{y} = \frac{1}{2}$$

$$\frac{\text{Moles of O in } Ca_3(PO_4)_2}{\text{Moles of O in } H_3PO_3} = \frac{8x}{3y} = \frac{8}{3} \times \frac{1}{2} = \frac{4}{3}$$

18. $$\frac{t_{3/4}}{t_{1/2}} = \frac{4^{(n-1)} - 1}{2^{n-1} - 1}$$

$$\Rightarrow \frac{t_{3/4}}{t_{1/2}} = \frac{\left(2^{n-1} - 1\right)\left(2^{n-1} + 1\right)}{\left(2^{n-1} - 1\right)} = \left(2^{n-1} + 1\right)$$

$\therefore\ t_{3/4} = t_{1/2}(2^{n-1} + 1)$

19. SF_4, I_3^- and PCl_5 have sp^3d hybridization at central atom, however, however $SbCl_5^{2-}$ has sp^3d^2 hybridization.

20. Nitro group is present at para position. (–R effect, which increase acidic strength)

24. $$K_a = C\alpha^2 = 0.02 \times \frac{10}{100} \times \frac{10}{100} = 2 \times 10^{-4}$$

25. $$\sqrt{\frac{\alpha_1}{\alpha_2}} = \sqrt{\frac{K_{a_1}}{K_{a_2}}}$$

26. $C \rightarrow B$ part in the graph represents fusion at a constant temperature of 600 K and for 40 min.
Time of fusion = 45 – 5 = 40 min

So, total heated absorbed during the fusion = $\frac{0.4\times10^3\times40}{2}\text{Jmol}^{-1}$

$\therefore\ \Delta S_{fusion} = \frac{0.4\times10^3\times40}{2\times600} = 13.33\,\text{Jmol}^{-1}\text{K}^{-1}$

Hence (B) is correct.

27. $\Delta H = \Delta U + \Delta(PV) = \Delta U + P_2V_2 - P_1V_1$

We cannot use, $\Delta H = \Delta U + \Delta nRT$ as $PV \neq nRT$ since gases deviate from ideal behaviour.
$\therefore\ W = P_2V_2 - P_1V_1 = 40 \times 1 - 70 \times 1 = -30$ lit atm $= -3$ kJ
Also, $-560 = \Delta U - 3.0$
$\Delta U = -557$ kJ

28. Higher the concentration of common ion, the smaller is the solubility.

29. $\frac{C_2}{C_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{1200}{300}} = 2$

$C_2 = 2C_1 = 2 \times 0.3\ \text{ms}^{-1} = 0.6\ \text{ms}^{-1}$

30. Since melting point of K is least among all, so heat released in the reaction immediately melts its surface and catches fire.
Hence (C) is correct.

Mathematics

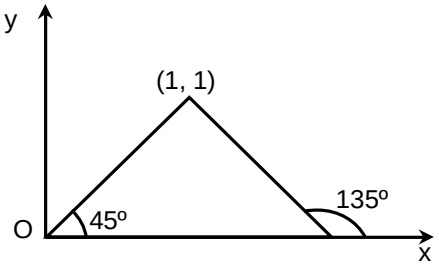
PART – III

SECTION – A

1. $l - 2m + n = 0$ ($\because$ l, m, n in A.P.)
compare with $lx + my + n = 0$

2. Slope of the lines are 1 & – 1
Hence required equation are
$(y - 1) = \pm 1 (x - 1)$

$x = 1, y = -2.$



3. Centre = (–2, 2), radius = 2

5. For centre of F(x, y) = 0 solve $\frac{df}{dx} = 0$ & $\frac{df}{dy} = 0$

6. Given parabola is $(x + 2)^2 = -2\left(y - \frac{11}{2}\right)$

7. $\frac{2b^2}{a} = b \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{3}}{2}$

10. Focal distance of any point P(x, y) on the ellipse is equal to (a + ex) : so 'a + ae cos θ' ($\because$ x = a cos θ).

11. Centre of both = (0, 0)
vertices of both = ($\pm$ 5, 0)

12. $\frac{1}{x} = \sqrt{7 - 4\sqrt{3}} \;\; \therefore \;\; x + \frac{1}{x} = \sqrt{7 + 4\sqrt{3}} + \sqrt{7 - 4\sqrt{3}} = \left(\sqrt{3} + 2\right) + \left(2 - \sqrt{3}\right) = 4$

13. $D = 25(l + m)^2 + 8(l - m)^2 > 0$

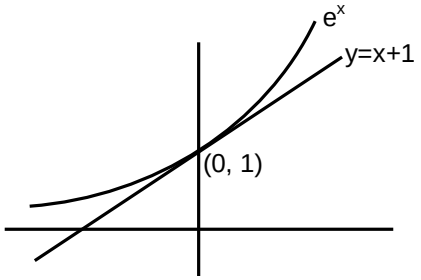
14. $e^x = x + 1 \Rightarrow 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \infty = x + 1$

i.e., $\frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots \frac{x^n}{n!} + \ldots = 0$

$\therefore \quad x^2 = x^3 = x^4 = \ldots = 0$

hence x = 0 only one real root

Alternate by graph, we see that
$y = x + 1$ touches $y = e^x$ at (0, 1) so only one real root.



15. Required number = $\frac{n\times p}{4}(n+1)(p+1)-\sum_{r=1}^{n}(n+1-r)(p+1-r)$,
where the given rectangle is of size $n \times p$ $(n < p)$
$n = 6, p = 9$.
$\therefore$ Number = $\frac{9\times 6}{4}(9+1)(6+1)-\sum_{r=1}^{6}(7-r)(10-r)$
$= 945 - \sum_{r=1}^{6}(70-17r+r^2)$
$= 945 - 154 = 791.$

16. Let us fix 3 at unit place.
$\therefore$ Number of possible numbers is = 3! = 6.
$\therefore$ Each number can make to occur at unit place 6 times.
$\therefore$ Required sum = $6(3 + 4 + 5 + 6) = 6 \times 18 = 108$.

17. $(1 + 3x + 3x^2 + x^3)^6 = [(1 + x)^3]^6 = (1 + x)^{18}$
$\therefore$ Middle term = $\left(\frac{18}{2}+1\right)^{th}$ term i.e., 10th term.

18. $\frac{1}{1-2x+x^2} = (1-x)^{-2} = 1+2x+3x^2+\cdots$
$= 1 + a_1x + a_2x^2 + \ldots$
$\Rightarrow$ $a_r = r + 1$.

19. Coefficient of $x^{17} = -(1 + 2 + 3 + \cdots + 18)$
$= -\frac{18}{2}(1+18) = -9\times 19 = -171.$

20. Let $\frac{a}{|z_2-z_3|}=\frac{b}{|z_3-z_1|}=\frac{c}{|z_1-z_2|} = \lambda$ (say)
$\Rightarrow$ $a = \lambda|z_2 - z_3|, b = \lambda|z_3 - z_1|, c = \lambda|z_1 - z_2|$
$a^2 = \lambda^2|z_2 - z_3|^2 = \lambda^2(z_2 - z_3)(\bar{z}_2 - \bar{z}_3)$
$\therefore$ $\frac{a^2}{(z_2-z_3)} = \lambda^2(\bar{z}_2 - \bar{z}_3)$
Similarly $\frac{b^2}{(z_3-z_1)} = \lambda^2(\bar{z}_3 - \bar{z}_1)$ and $\frac{c^2}{(z_1-z_2)} = \lambda^2(\bar{z}_1 - \bar{z}_2)$
$\therefore$ $\frac{a^2}{(z_2-z_3)}+\frac{b^2}{(z_3-z_1)}+\frac{c^2}{(z_1-z_2)} = 0$

21. Remainder when f(x) is divided by $x - \alpha$ is $f(\alpha)$.
Let the remainder when f(z) is divided by $z^2 + 1$ is $ax + b$.
[remainder will be in general a linear expression as $z^2 + 1$ is quadratic]
Then $f(z) = (z^2 + 1)\, g(z) + (az + b)$ …(1)
Now $f(i) = i \Rightarrow ai + b = i$ …(2)
and $f(-i) = 1 + i \Rightarrow -ai + b = 1 + i$ …(3)

Solving (2) and (3), we get $b = \frac{1}{2} + i$ and $a = -\frac{1}{2i} = \frac{i}{2}$

Hence, the required remainder is $\frac{1}{2}iz + \frac{1}{2} + i$.

22. $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$

$\Rightarrow \quad \frac{1}{z_1} - \frac{1}{z_2} = \frac{1}{z_3} - \frac{1}{z_1} \quad \Rightarrow \quad \frac{z_2 - z_1}{z_1 z_2} = \frac{z_1 - z_2}{z_1 z_3}$

$\Rightarrow \quad \frac{z_2 - z_1}{z_3 - z_1} = -\frac{z_2}{z_3} \quad \Rightarrow \quad \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arg\left(-\frac{z_2}{z_3}\right) = \pi - \arg\left(\frac{z_2}{z_3}\right)$

$\Rightarrow \quad \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \pi - \arg\left(\frac{z_2 - 0}{z_3 - 0}\right)$

$\Rightarrow \quad \angle CAB = \pi - \angle COB$

$\Rightarrow \quad \angle CAB + \angle COB = \pi \Rightarrow$ the points O, A, B and C are concyclic.

23. Let $\alpha = x + iy$ and $\beta = x - iy$ and a, b > 0

Since $|\alpha| = |\beta| \quad \therefore \frac{\alpha + \beta}{\alpha - \beta} = \frac{2x}{2iy} = -\frac{ix}{y}$

Again, if $\alpha = x - iy$ and $\beta = -x - iy$

Then $\frac{\alpha + \beta}{\alpha - \beta} = \frac{-2iy}{2x} = -\frac{iy}{x}$

24. As x, y, z are in G.P. $\Rightarrow y^2 = xz$
$\Rightarrow \ln(y)^2 = \ln(xz)$
$\Rightarrow 2\ln(y) = \ln(x) + \ln(z)$
$\Rightarrow 2\{1 + \ln(y)\} = \{1 + \ln(x)\} + \{1 + \ln(z)\}$
$\Rightarrow 1 + \ln(x), 1 + \ln(y), 1 + \ln(z)$ are in A.P.
$\Rightarrow \frac{1}{1+\ln(x)}, \frac{1}{1+\ln(y)}, \frac{1}{1+\ln(z)}$ are in H.P.

25. We can write S as
$S = (1 - 2)(1 + 2) + (3 - 4)(3 + 4) + \ldots + (2001 - 2002)(2001 + 2002) + 2003^2$
$= -[1 + 2 + 3 + 4 + \ldots + 2002] + 2003^2$
$= -\frac{1}{2}(2002)(2003) + 2003^2 = 2007006.$

26. Let two positive numbers be a and b. Then x = (a + b)/ 2.
Also a, y, z, b are in G.P. If r is the common ratio of this G.P., then $b = ar^3 \Rightarrow r = (b/a)^{1/3}$
We have $\frac{y^3 + z^3}{xyz} = \frac{a^3r^3 + a^3r^6}{x(ar)(ar^2)} = \frac{a(1+r^3)}{x} = \frac{a+b}{(a+b)/2} = 2.$

27. $f(2003) = f(2003 + 0) = f[2003(0)] = f(0) \Rightarrow f(0) = 2003.$
Also, $f(-2003) = f(-2003 + 0) = f(0) = 2003$
It can also be seen that f is a constant function.

28. $a^2 + 4b^2 = 12ab$

$\Rightarrow \quad a^2 + 4b^2 + 4ab = 16ab$

$\Rightarrow \quad 2\log(a + 2b) = \log 16 + \log a + \log b$

$\Rightarrow \quad \log(a+2b) = \frac{1}{2}[4\log 2 + \log a + \log b]$

$= 2\log 2 + \frac{1}{2}(\log a + \log b)$

29. The given expansion is valid only for $-1 < x \leq 1$.

30. $\tan 3\theta = \tan\left(\frac{\pi}{2} - \theta\right)$

$\therefore \; 3\theta = n\pi + \left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow \; 4\theta = (2n+1)\frac{\pi}{2} \Rightarrow \theta = (2n+1)\frac{\pi}{8}$